

**Experiential Learning
Project Assessment-I
on
“ Python Snake Game”
under the course
Python Programming Language (N-PCCCM304P)**

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**Session:2026-27 (ODD)
Department of CSE(AI&ML)**

**S. B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT AND
RESEARCH,NAGPUR**

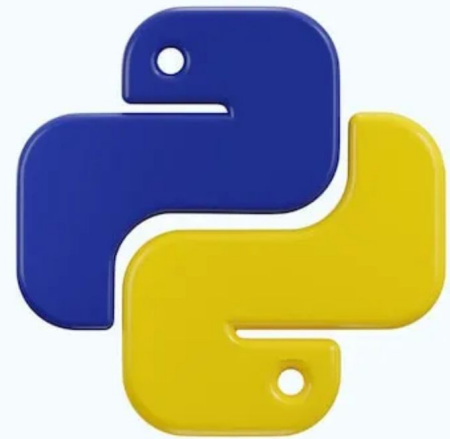
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Problem Statement :

Develop a **Python-based Snake Game** in which the player controls a snake that moves continuously on the screen to collect food. Each time the snake eats food, its length increases and the player's score is updated. The game ends if the snake collides with the screen boundaries or with its own body. The application should provide **smooth gameplay, score tracking, and options to restart or exit after the game is over.**

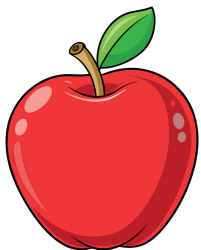
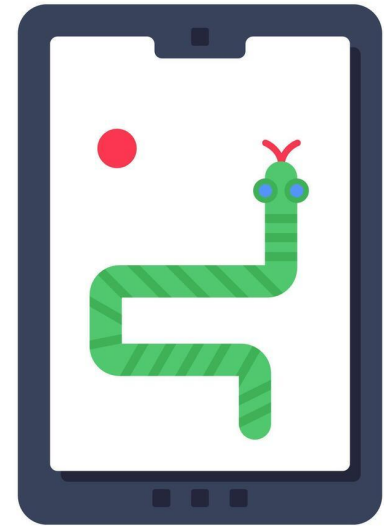


Score: **2** High Score: **2**



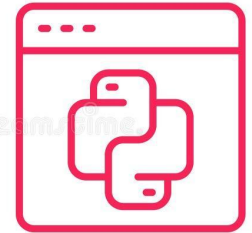
Objectives :-

1. To develop a functional Snake game using Python and Pygame.
2. To improve programming skills involving Python classes, lists, functions, loops, conditions, and event handling.
3. To develop practical Based learning by implementing real life situations
4. Understand logic used to build a game.



Algorithm:-

GAME OVER

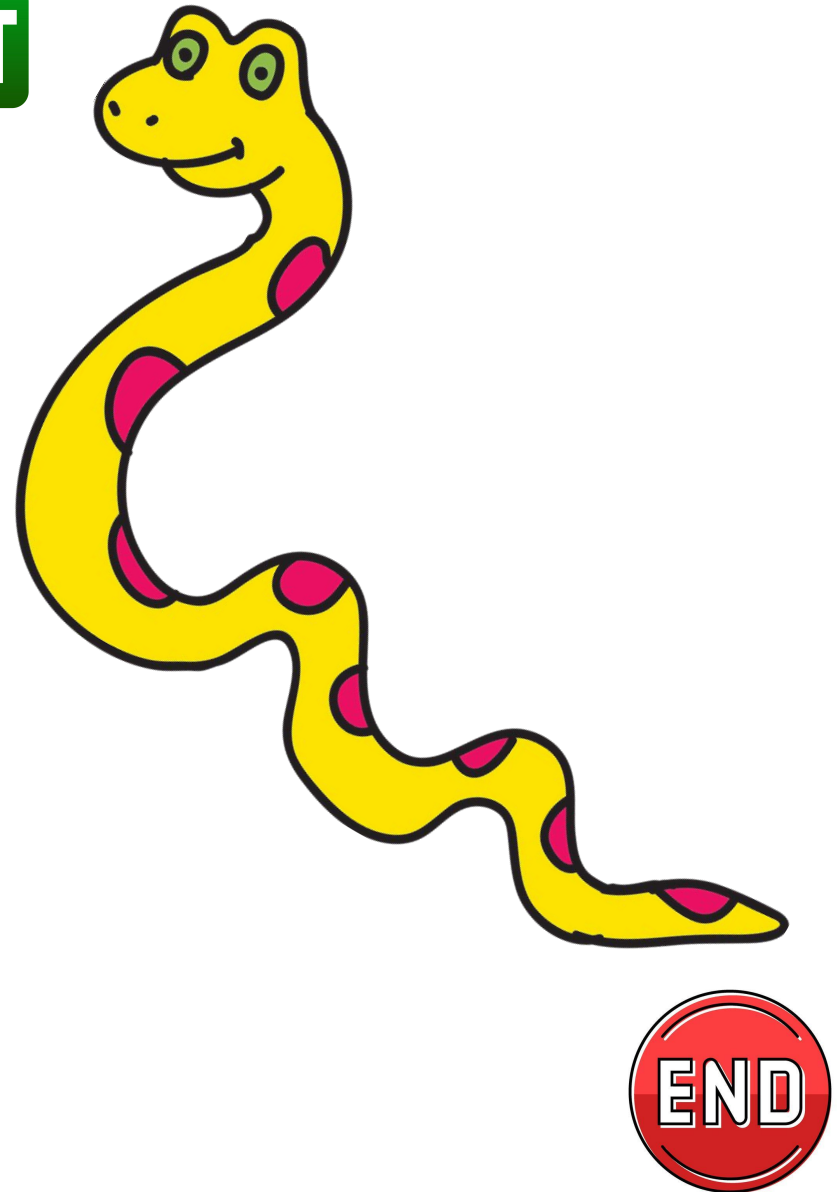
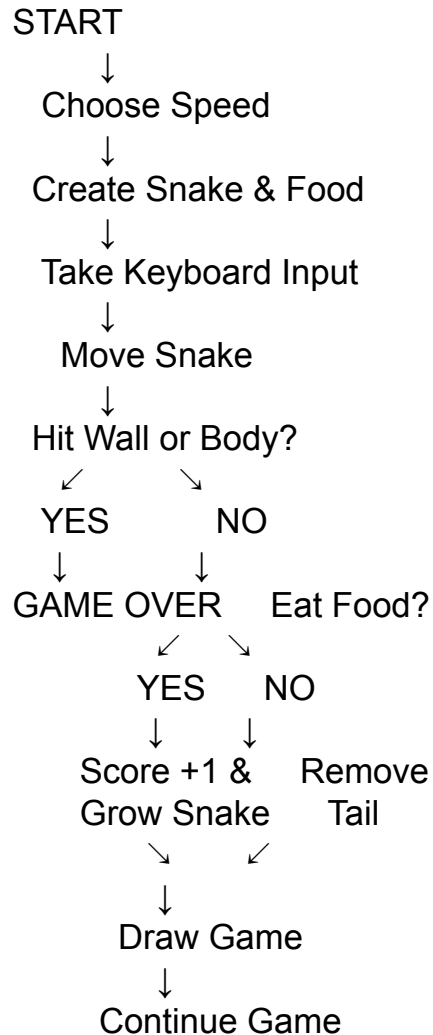


dreamstime

START GAME

1. Start the game and choose speed.
2. Create the snake and food.
3. Take keyboard input for movement.
4. Move the snake in the selected direction.
5. Check wall/body collision.
6. If food is eaten, increase score and snake length.
7. Generate new food.
8. Continue until collision occurs.
9. Display **Game Over** and allow restart or quit.

Flow Chart: **START**



Logic:-

Start → Create snake and food.

Input → Player chooses the snake's direction.

Move → Snake's head moves one grid cell.

Collision Check → Check wall or snake body.

Food Check → If snake eats food, score increases and snake grows.

New Food → Food appears at a random empty position.

Repeat → Continue moving until a collision occurs.

Game Over → Display final score and allow restart.



Thank You